



KAIROS

Kinetics of Aortic bioprosthesis Integrity, Risk-adapted echO Surveillance

καιρός, the right moment.

Dyania Health Hackathon 2026 · AVR Durability Challenge

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THE PROBLEM

The valve wears out quietly, and one schedule watches everyone

1 in 3

surgical bioprostheses showed haemodynamic deterioration over a decade, in a 1,387-patient cohort

10.8%

VARC-3 moderate or severe deterioration at five years after transcatheter implant, in 2,403 patients

29

of our own 117 patients carry an externally mounted pericardial valve, the Trifecta class. The FDA warned about it in 2023 and the maker withdrew it

- **Surveillance is fixed.** The same interval for a 42-year-old with a small externally mounted valve and an 84-year-old with a large one.
- **Attendance is not.** Five-year echo follow-up was missing in 30.4% of transcatheter and 35.7% of surgical patients in a low-risk trial.
- **Readings vary between sites,** and gradients depend on flow.

REOPERATIVE AVR, UK 1996-2019	MORTALITY
Elective	4.8%
Urgent	11.8%
Emergency	31.7%

An association across all reoperation causes, not a count of preventable deaths.

THE QUESTION

Nobody predicts. They detect.

Current practice asks whether a valve *is* failing. It never asks whether it *will*.

WHAT A CARDIOLOGIST DOES TODAY

Opens the echo, compares three numbers against the patient's own baseline. **Gradient risen 10 or more, now 20 or more** means moderate deterioration. Otherwise: nothing wrong, return in a year. A threshold, not a prediction.

WHAT WE ASK

Can patient, prosthesis and serial echocardiography predict adjudicated deterioration **beyond valve age and type**, and beyond that threshold rule? Guideline surveillance stays. The model only says **who needs looking at sooner**.

Structural deterioration, with death competing

ELEMENT	DEFINITION
Time zero	The patient's own reference echo, 30 to 180 days after implant. Not a fixed day 90
Primary	First adjudicated VARC-3 stage 2 or 3 haemodynamic deterioration attributable to SVD
Key secondary	Bioprosthetic valve failure. Deterioration is primary because failure is a treatment decision: undercounted, and too late to change surveillance
Competing	Death; index-valve replacement for a non-SVD cause
Intercurrent	Treated thrombosis resolves, so the patient returns to risk
Cohort	3,400 to 5,100 patients by Riley's criteria, counted in patients and events, never in landmark rows

TWO RULES THAT DECIDE VALIDITY

A day-90 prediction cannot use an echo taken on day 150.

Anchoring on the actual reference study removes the contradiction and the leakage with it.

The echo that establishes an endpoint is never a predictor of that endpoint. Without this rule the model looks brilliant and is worthless.

Real records in, synthetic cohorts out

INGESTED · PRIVATE, NEVER LEAVES THE MACHINE

202 43,550 5,807

notes, 117 patients lab rows medication rows

INGESTED · PUBLIC, 18 SOURCES, 567 MB, HASHED

Devices	FDA GUDID full release, openFDA MAUDE counts
Haemodynamics	ASE 2024 prosthetic valve reference tables; 10 FDA SSEDs
Outcomes	ClinicalTrials.gov, 10 trials, 755 measures; NOTION and Trifecta papers

Only 1 patient has gradients in two different years. Extraction works; training cannot. The model is developed on six synthetic scenarios, and every figure says so.

PREPARED

Device normalisation	42 canonical valve models, 75 free-text aliases, 256 ASE reference-value rows mapped to them
Valve passport	Model, generation, size, route and echo series from free text. Span-aware rules first; gpt-5.1 in-region behind a flag. Every field keeps its evidence span
Labs and medications	11,818 non-clinical lab rows removed; anticoagulant exposure timelines; warfarin INR time in range
Adjudication	VARC-3 staging against the patient's own reference echo, with an uncertain class; three extraction traps pinned by tests
Landmarks	Re-scored at every echo; leakage rules enforced, the endpoint echo never a predictor
Synthetic cohorts	Six scenarios from cited literature, a generator calibrated to a published-evidence registry, fixed seeds
Privacy	Aggregates suppressed below 5; a CI scan blocks any patient-level row from version control

One inference, three probabilities

EXTRACT Valve passport from free text, with evidence spans	ADJUDICATE VARC-3 stage against the reference study	PREDICT Cause-specific hazards: SVD, death, non-SVD replacement	UPDATE Re-scored at every echo; biomarker and anticoagulant modules	REPORT Three probabilities at 1, 3, 5 years, drivers, reliability
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MODELS

Served	Penalised cause-specific Cox (lifelines, elastic net), per route and cause, combined into cumulative incidence by piecewise-constant hazard integration
Challenger	Gradient-boosted survival (scikit-survival), nested tuning, same folds. Did not beat Cox; Cox retained
Comparator	The VARC-3 threshold rule, implemented independently of the model, for net benefit
Evaluation	IPCW Brier, calibration slope, time-dependent AUC, IPA, 200 patient-level bootstraps, support gates that abstain
Language model	Azure OpenAI gpt-5.1, Sweden Central only: extraction, adjudication proposals, a grounded patient summary whose numbers are checked against the facts

TECHNOLOGY

Python 3.11	pandas · SciPy	lifelines	scikit-survival
pydantic	FastAPI extract, predict	Streamlit demo	
Azure Container Apps + Jobs	Container Registry	PostgreSQL	
Blob Storage	Key Vault	App Insights	Bicep IaC
281 tests · ruff · privacy scan			

WHY THREE AND NOT ONE

One minus the deterioration risk is **not** the chance of being alive with a working valve: it includes everyone who died first. In a population this old, the complement overstates durability.

Beating valve age proves nothing. Beat the rule.

One patient, 58, bicuspid valve, four years after a surgical implant.

MEASURE	DAY 90	YEAR 4
Mean gradient, mmHg	11	19
Dimensionless index	0.48	0.37
Effective orifice area, cm ²	1.7	1.25

Rose 8 mmHg. Stage 2 needs a rise of 10 to reach 20. He has crossed no line.

THE GAP

The rule says *nothing is wrong, return in a year.*

The model reads the trajectory and brings the assessment forward to six months.

Net benefit against that rule is the primary clinical comparison. Illustrative values; the model shown is trained on synthetic scenarios.

Gains where information exists. Losses reported.

IPCW BRIER AT 5 YEARS	REF	CORE	+MODULES	EVENTS
Gradual stenotic	.0559	.0472	.0485	198
Biomarker, meaningful	.0524	.0476	.0468	178
Biomarker, absent	.0490	.0478	.0475	159
Irregular surveillance	.0639	.0602	.0584	189
Regurgitant, abrupt	.0776	.0782	.0790	267
High competing mortality	.0041	.0042	.0039	3

Lower is better. Reference is valve age and type alone, the weak bar. The guideline threshold rule, run on the same 10,701 landmarks, returned an evaluable verdict on **0**: it abstains without a complete severity assessment. Net benefit therefore remains unmeasured.

WHERE IT WORKS

Gradual stenotic: **AUC 0.80 (0.77–0.83)**, calibration slope **0.88**, IPA **0.21**. Without serial echo, Brier rises to **.0575**, worse than the reference. Updating is the whole gain.

WHERE THE MODULES STAY QUIET

The biomarker block helps when the scenario carries information and does nothing when it does not, as specified in advance.

GRADIENT BOOSTING, THE PRE-SPECIFIED CHALLENGER

Same rows, same folds, 200 patient bootstraps. Brier difference **+5.7e-4** [–4.0e-6, +1.1e-3]. It failed its own promotion rule: **Cox retained**.

WHERE IT LOSES, SAID FIRST

On **abrupt regurgitant failure** the reference beats us: no gradient trend precedes it, and calibration slope falls to 0.67.

Note in, decision support out, running on Azure

FOR EACH PATIENT, AT EVERY ECHO

- 1** Valve passport read from the note: model, generation, size, route, echo series, each value linked to its source text
- 2** VARC-3 stage against the patient's own reference study, with an uncertain class
- 3** Three probabilities at 1, 3 and 5 years: deterioration before death, death before deterioration, alive with the valve intact
- 4** Drivers and a reliability statement with reason codes, abstaining where support is thin
- 5** Patient Summary: a plain-language narrative, every number checked against the facts
- 6** The guideline schedule, unchanged, shown beside the model so it never replaces it

LIVE PROTOTYPE

kairos-demo.blackground-92784e60
.swedencentral.azurecontainerapps.io

Patient Summary

Patient inputs, all editable

Prediction

Compare families

Risk over time

What-if: next echo

Fill from a note

BEHIND IT

Extract and predict APIs with OpenAPI contracts; jobs that generate, train, evaluate and publish models with rollback; one-command deploy.

REPRODUCIBLE

make test && make quick

Extraction, six scenarios, training and evaluation from a clean checkout.

An order we will not skip

1 · RETROSPECTIVE

Development in cohorts with adjudicated endpoints

2 · FREEZE

Temporal and external validation, untouched centre

3 · SILENT

Prospective, no clinical action on the output

4 · ONLY THEN

A trial of whether intervals can safely change

An observational model cannot show that looking *less* often is safe.
So we never say it. It says *who to look at sooner*.